

Jacob Berkowitz

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EXPERIENCE

Amazon (AWS)

March 2024 - March 2026

Software Development Engineer I

Seattle, WA

- Design and maintain a three-tier PKI infrastructure with automated subordinate certificate rotation using Lambda functions for secure signing workflows across 3 AWS partitions, adhering to ITAR regulations and ensuring high security for sensitive customer data in GovCloud.
- Restored and reengineered the Apple signing process for the Notation plugin by repairing a broken signing workflow, enhancing code quality, incorporating testing, implementing a ToD worker-based signing system, and refining the 32-step modeled change management procedure.
- Build and manage 7+ scalable, fault-tolerant CI/CD pipelines using AWS CDK and CloudFormation with automated rollback mechanisms, Hydra-based approval workflows, accelerated releases, and region expansion, enabling daily deployments to 20+ regions worldwide and reducing deployment time by 33%.
- Conducted load testing and optimized resource usage to scale backend systems from 800 → 2000 TPS for large-file signing operations; added JARSigner verification for 20+ self-signed certificates, enhanced monitoring across 12+ PCAs for improved security visibility, assisted in resolving service issues during an AWS outage, and led a Correction of Errors review in response to a Sev-2, contributing to several others.
- Create and own the process to manually rotate 4+ expiring Intermediate PCAs in Signer's PKI infrastructure; partnering with another team to transition PCA ownership and implement a centrally managed Root and Intermediate CA system, eliminating future rotation overhead for Signer.

Amazon (AWS)

June 2023 - September 2023

Software Development Engineer Intern

Seattle, WA

- Contributed to the AWS Signer team on the SWIGs Operations Plane project.
- Designed and delivered a new operations plane in Python and TypeScript to streamline on-call workflows, enabling the execution of 30+ tools through a unified interface.
- Extended an internal operations tool execution framework, originally designed for EC2, to support ECS with Fargate, improving scalability and deployment flexibility.
- Refactored the framework into a standalone package for cleaner separation of concerns and easier maintainability, and built a custom Docker container image to optimize Fargate deployments.
- Created a scalable CDK pipeline for deploying CloudFormation resources and container services across 20 AWS regions.

Amazon (AWS)

May 2022 - August 2022

Software Development Engineer Intern

Seattle, WA

- Collaborated with the AWS Signer team on the Revocation Data Backfill for New Region Builds project.
- Built an AWS CDK pipeline to automate cross-region and cross-account transfer of a 100-GB DynamoDB table, enabling scalable and reliable data replication.
- Implemented a Java package leveraging parallel scans for high-throughput data reads and batch writes to migrate large-scale table data efficiently.

EDUCATION

University of Michigan, Ann Arbor

May 2023

Bachelor of Science - Computer Engineering with Cum Laude

SKILLS

Languages: Python, Java, Typescript, Ruby, JSON, YAML, C++, C, Verilog, MATLAB, SQL, Arduino

Technologies/Frameworks: AWS, CDK, Git, Docker, Windows/OSX/Linux, Cloudformation, CI/CD Pipelines, Unit/Integration Testing, PKI Systems